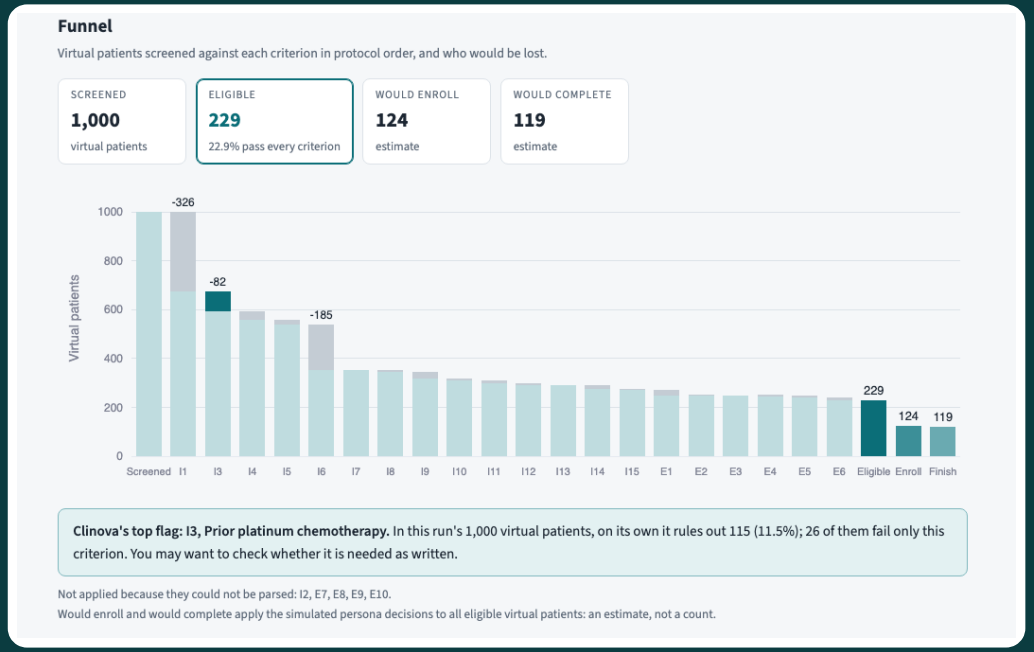
# Clinova

Test-fly your NSCLC trial on thousands of virtual patients before enrolling a real one




Every prediction is graded against how real ClinicalTrials.gov trials turned out

https://github.com/RaghulKS/Clinova

THE PROBLEM

# Most trial delays are written into the protocol before the first patient is screened

**57%**

of protocols need a substantial amendment

45% of those are avoidable

**$535K**

median cost of one Phase III amendment

$141K in Phase II, plus about 3 months

**34.5%**

of cancer-trial terminations: low accrual, the top reason

22.74% of 8,687 US cancer trials terminated

Clinova pulls the two levers the FDA cost study names: looser enrollment rules and simpler protocols

Sources: docs/ARCHITECTURE.md

WHY NOW

# Public version history and LLM parsing make protocol risk checkable today

1

## History is public

ClinicalTrials.gov keeps every registered version, so a widened rule is a checkable label

2

## LLMs read eligibility

Gemini turns free-text criteria into rules that code can check on virtual patients

3

## Design is the AI gap

Tufts CSDD 2024: full AI adoption is lowest in trial design and planning

Sources: docs/ARCHITECTURE.md

WHAT CLINOVA DOES

# Clinova test-flies a protocol in five steps, from draft text to a graded risk

1

## Read

Draft text or NCT ID to coded rules and visits

2

## Simulate

Virtual NSCLC cohort, screened in code; persona agents decide

3

## Calibrate

LightGBM on real trials, explained by SHAP

4

## Optimize

Test edits; see trade-offs with ranges

5

## Learn

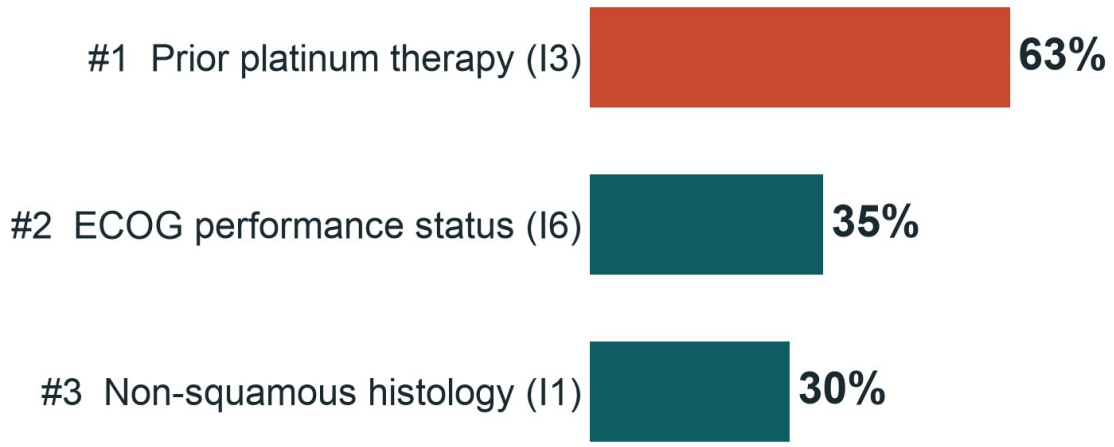
Log every prediction, score it when the trial ends

Sources: docs/ARCHITECTURE.md

BACKTEST ON A REAL TERMINATED TRIAL

# Given only its first protocol, Clinova's flag #1 was the rule the sponsor later removed

Top 3 flags: share of virtual patients each rule excludes on its own




Coral: the rule the sponsor later removed

## NCT03232892

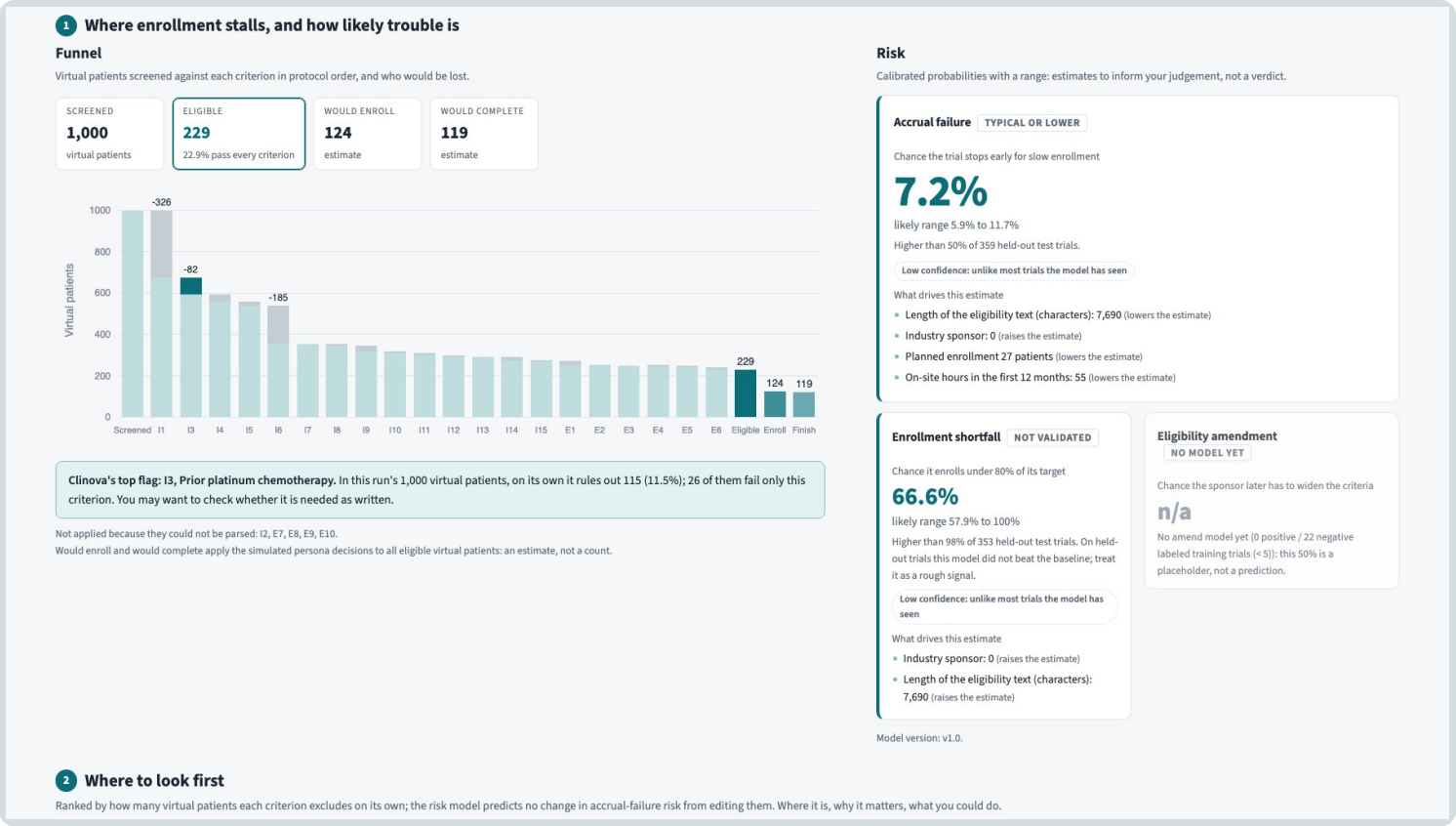
Terminated 2019-09-11: “Low Accrual”

- Flag #1: prior platinum therapy (I3), the rule the sponsor later removed; alone it excludes 63%
- Flags rank rules by how many virtual patients each excludes; the risk model does not read criteria
- Honest miss: predicted accrual-failure risk 7% (range 6% to 12%), the 50th percentile of test trials

Sources: docs/demo_trial.md, docs/demo_results/NCT03232892.json

LIVE DEMO

# The app shows where enrollment stalls and how likely trouble is, in plain words




- Funnel: who is excluded by which rule, who enrolls, who completes
- Risk: probability with a range, its rank among test trials and plain-word drivers
- Labels say “not validated” or “no model yet” where the evidence is thin

Sources: docs/screens/03-funnel-risk.png

OPTIMIZER

# The optimizer proposes three trade-offs with ranges; the author decides which, if any

## Fastest enrollment

Widen histology rule (I1)
Lift ECOG limit (I6)
CrCl (I14): ≥ 50 to ≥ 20 mL/min

| | |
|---|---|
| Enrollment speed | **+101%** +98% to +137% |
| Dropout | **+4.4 pts** +4.4 pts to +6.8 pts |
| Diversity index | **+0.039** |

## Fewest dropouts

Drop bilirubin rule (I11)
Drop histology rule (I1)
Drop cardiac exclusion (E6)

| | |
|---|---|
| Enrollment speed | **+53%** +51% to +66% |
| Dropout | **-2.0 pts** -2.0 pts to +0.9 pts |
| Diversity index | **+0.019** |

## Balanced

ECOG (I6): ≤ 1 to ≤ 2
Drop histology rule (I1)
Drop CrCl (I14)

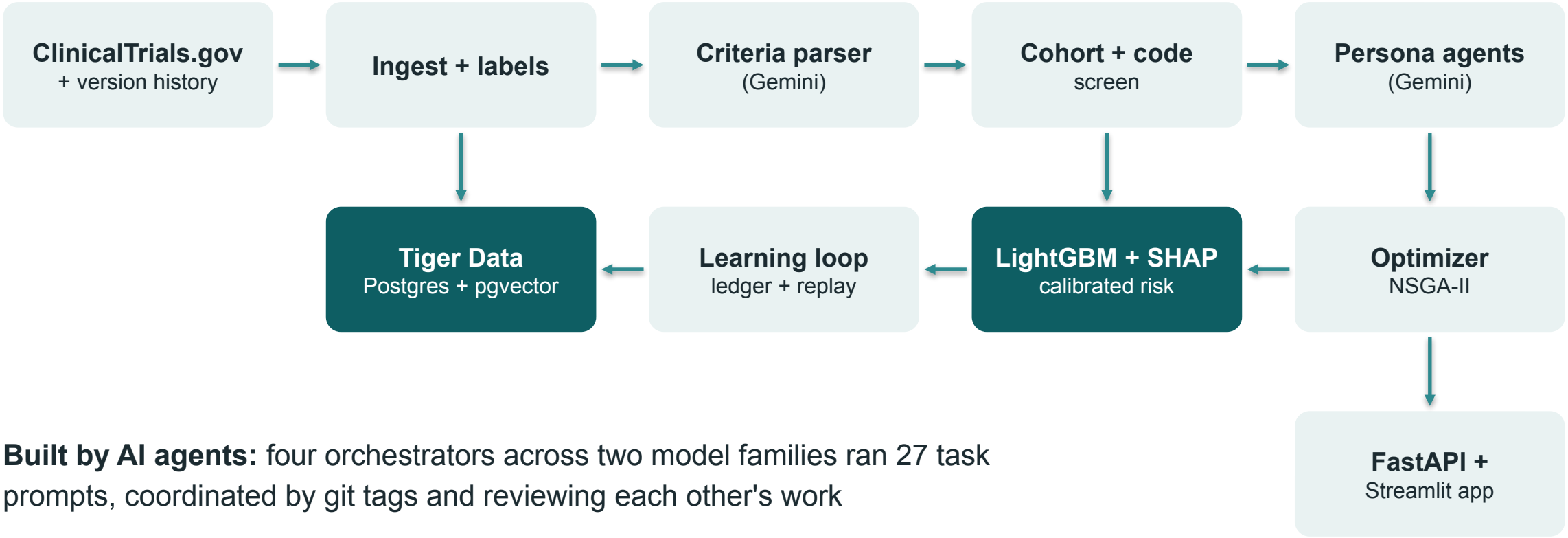
| | |
|---|---|
| Enrollment speed | **+95%** +90% to +128% |
| Dropout | **+3.6 pts** +3.6 pts to +5.0 pts |
| Diversity index | **+0.037** |

Ranges: 5 cohort seeds. No dollar value yet (no amendment labels). Authors lock rules that define the trial's population

Sources: docs/demo_results/NCT03232892_optimize.json

ARCHITECTURE

# Small, tested modules around one Tiger Data store, built entirely by AI agents

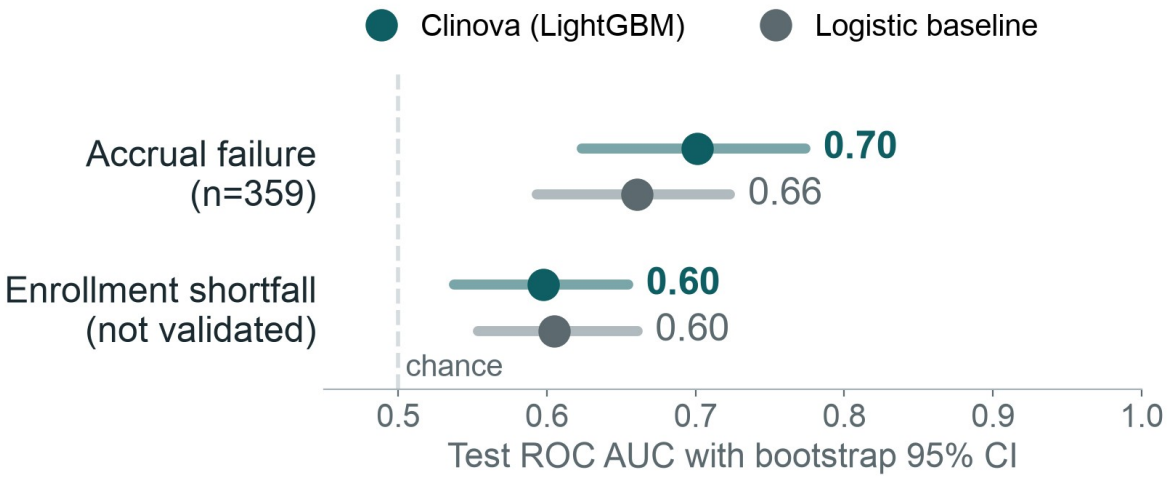


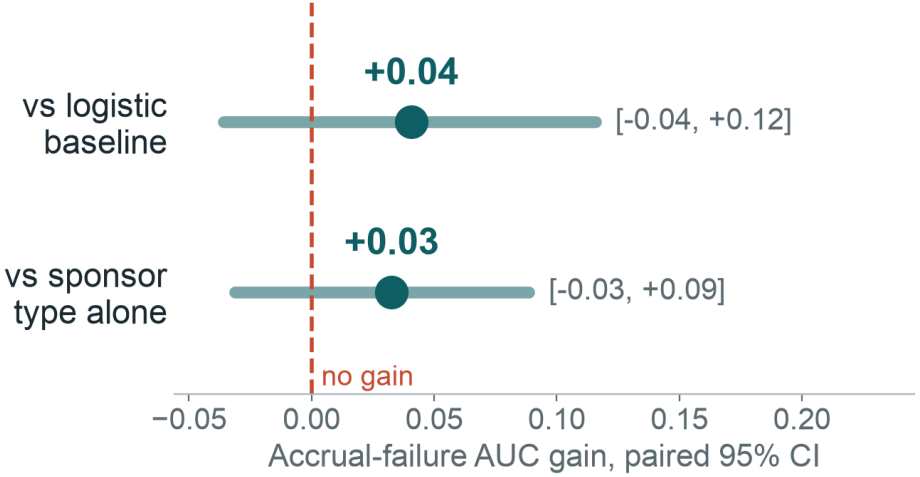

**Built by AI agents:** four orchestrators across two model families ran 27 task prompts, coordinated by git tags and reviewing each other's work

Sources: docs/ARCHITECTURE.md, orchestration/tasks/

VALIDATION ON TRIALS THE MODEL NEVER SAW

# On 359 unseen trials, accrual-failure AUC is 0.70: above chance, not yet shown to beat a simple baseline



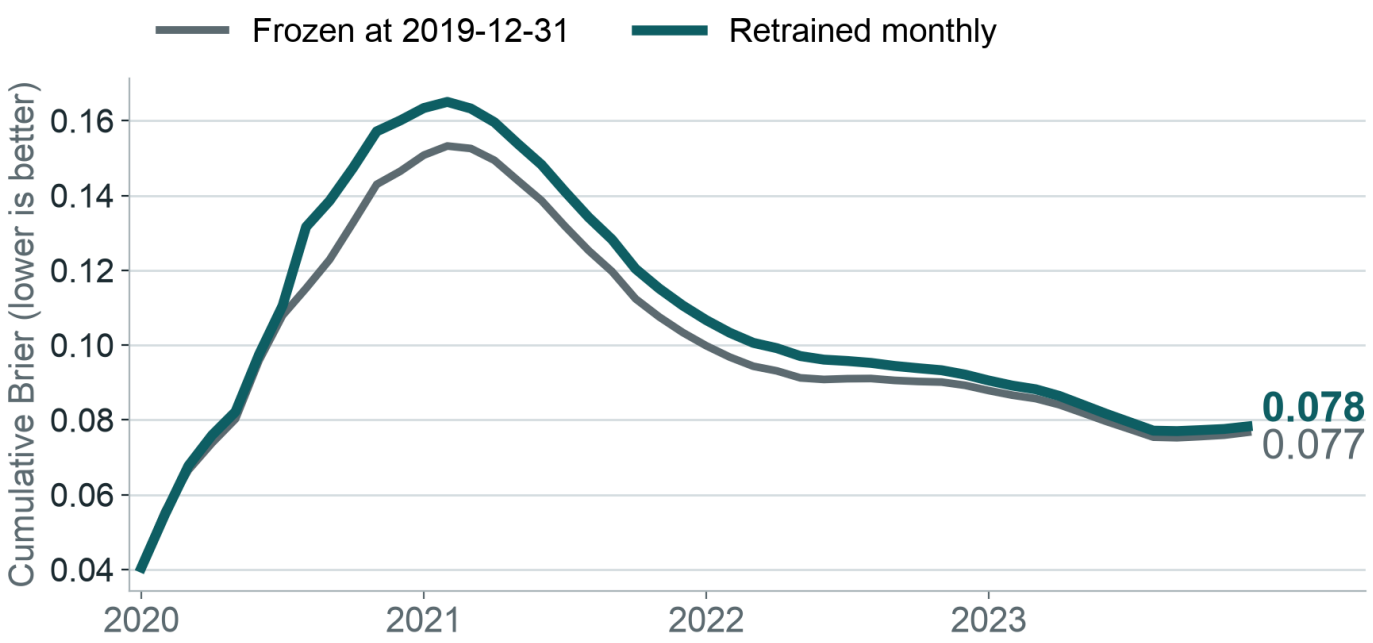




- Accrual failure: AUC 0.70 (95% CI 0.62–0.77) vs logistic baseline 0.66 (0.59–0.72)
- Paired AUC gain +0.04 (95% CI −0.04 to +0.12): the interval includes zero
- Brier 0.083 vs 0.086 for predicting the base rate; hit rate and simulation benefit not computable yet

Sources: docs/metrics.json, docs/metrics_detail.json

LEARNING LOOP

# In a 2020–2023 replay, monthly retraining did not beat the frozen model (Brier 0.078 vs 0.077)




- Replay 2020-01 to 2023-12: score each month, then learn its outcomes
- Pooled Brier over 1,736 overlapping windows: retrained 0.078, frozen 0.077
- Every live prediction is logged in Tiger Data and scored when the trial ends

Sources: docs/replay.json

LIMITATIONS AND RESPONSIBLE USE

# We show what the evidence can and cannot support

### Not shown to beat simple baselines

Accrual AUC is above chance, but its paired gain over a logistic baseline includes zero

### Amendment labels are missing

Version histories are not loaded yet, so hit rate and amendment risk are unmeasured

### Personas are synthetic

Clinical traits come from published distributions; too few simulated trials to test their benefit

### Scores are correlational

They rank protocols against similar trials; they do not show that a rule caused a failure

**Clinova advises; the author decides. No real patient data. Every decision is logged.**

Sources: docs/gate2_report.md, docs/metrics.json

ROADMAP

# Next: more indications, site-level simulation and structured protocol input

1. Lymphoma, ophthalmology and immunology
2. Site-level enrollment simulation
3. Persona model tuned on expert feedback
4. Structured protocol input (ICH M11)
5. Full GxP validation package

Run it yourself

**github.com/RaghulKS/Clinova**

MIT license. docker compose up, then open the app; demo mode needs no API keys.

Sources: docs/ARCHITECTURE.md